Supplementary Information

Cognitive architecture moderates emotion–cognition relationships: cross-domain evidence from meta-analytic reanalysis

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Table 1: *

Supplementary Table 1. Specific falsifiable predictions derived from the DLN framework for emotion–cognition relationships. Each prediction names the specific construct, proposed measure, and expected direction by DLN stage. Predictions are numbered to facilitate tracking across future empirical tests.

#	Construct	Prediction	Measure(s)
S1	Emotional granularity	Granularity increases across DLN stages: dot < linear < network. Dot-stage individuals show undifferentiated affect; linear-stage individuals show basic category distinctions; network-stage individuals show fine-grained differentiation.	Levels of Emotional Awareness Scale (LEASLane et al. [1990]); experience sampling affect grids
S2	Alexithymia (curvilinear)	TAS-20 scores show a curvilinear relationship with DLN stage: *highest* for linear-stage individuals (active suppression), moderate for dot-stage (emotional opacity without suppression), *lowest* for network-stage (integration).	Toronto Alexithymia Scale (TAS-20Bagby et al. [1994])
S3	Decision quality under load	Dot: impaired across load levels. Linear: intact under low load, sharp decline under high load. Network: gradual, proportional decline. The stage × load interaction is the key test.	Iowa Gambling Task variants; Balloon Analogue Risk Task under affect induction
S4	Dangerous-middle	Individuals with relational emotional knowledge but low metacognitive monitoring exhibit *worse* outcomes under high affective load than linear-stage compartmentalisers. This reversal is not predicted by monotonic stage models. *Supported:* Hoyt et al. (2024) show linear-coded domains produce significantly worse outcomes than dot-coded domains ($b = -0.16$, 95% CI $[-0.30, -0.01]$).	LEAS (relational knowledge); structural-revision tasks (monitoring); high-load decision paradigms
S5	Implicit–explicit gaps	Implicit–explicit attitude discrepancies are largest for linear-stage individuals (systematic suppression) and smallest for network-stage individuals (integration aligns implicit and explicit). Dot-stage shows moderate discrepancy (inconsistency, not systematic suppression).	Implicit Association Test (IATGreenwald et al. [1998]); explicit self-report attitude scales
S6	Recovery after disruption	Recovery time after emotional disruption is slowest for linear-stage (re-establishing suppression), fastest for network-stage (integrating experience), intermediate for dot-stage (reactive dissipation).	Affect recovery paradigms; physiological return-to-baseline measures

Supplementary Table 2: Coding sensitivity audit

To assess robustness to coding ambiguity, we identified the most debatable stage assignments in each dataset and reran all analyses under every plausible alternative coding. For each scenario, we report the residual $\hat{\tau}^2$, percentage reduction from baseline, and whether the qualitative sign pattern is preserved. Across all 47 scenarios (16 DLN-stage and 31 Greenwald topology) the qualitative pattern was preserved in every case (47/47, 100%). Among the 16 DLN-stage scenarios, $\hat{\tau}^2$ reduction exceeded 50% in 12 cases (75%).

Debatable assignments. *Webb et al. (2012):* acceptance (network vs. linear), concentration (dot vs. linear), situation modification (linear vs. network); $2^3 = 8$ scenarios. *Interoception:* BAQ (network vs. dot), EDI-IAw (linear vs. dot); $2^2 = 4$ scenarios. *Hoyt et al. (2024):* behavioural (dot vs. linear), physical health (dot vs. linear); $2^2 = 4$ scenarios. *Greenwald et al. (2009):* drugs/social (linear vs. linear-plus), clinical/stigma (linear vs. linear-plus), race deliberative (linear-plus vs. linear), gender (linear-plus vs. linear), other intergroup (linear vs. linear-plus); 31 scenarios.

Table 2: *

Supplementary Table 2. Coding sensitivity audit. For each dataset the baseline (intercept-only) $\hat{\tau}^2$ is shown once; each row gives the residual $\hat{\tau}^2$ after adding the DLN stage moderator under one alternative coding, the percentage reduction, and whether the expected qualitative sign pattern is preserved. Primary coding is marked with †. Pattern key: net>lin>dot (monotonic gradient); sign reversal (lin+, net−); V-shape (lin<dot, lin<net).

Dataset	Recoded assignments	$\hat{\tau}^2_{\text{base}}$	$\hat{\tau}^2_{\text{mod}}$	% Red.	Pattern
Webb ($k = 10$)	Primary coding†	0.0350	0.0105	69.9	✓
	sit. mod. → network		0.0117	66.7	✓
	conc. → linear		0.0287	18.2	✓
	conc. → linear; sit. mod. → network		0.0299	14.6	✓
	accept. → linear		0.0094	73.1	✓
	accept. → linear; sit. mod. → network		0.0104	70.4	✓
	accept. → linear; conc. → linear		0.0260	25.8	✓
	accept. → linear; conc. → linear; sit. mod. → network		0.0274	21.9	✓
Intero-ception ($k = 8$)	Primary coding†	0.1339	0.0206	84.6	✓
	EDI-IAw → dot		0.0246	81.6	✓
	BAQ → dot		0.0178	86.7	✓
	BAQ → dot; EDI-IAw → dot		0.0263	80.3	✓
Hoyt ($k = 8$)	Primary coding†	0.0382	0.0000	100.0	✓
	phys. health → linear		0.0023	94.1	✓
	behav. → linear		0.0000	100.0	✓
	behav. → linear; phys. health → linear		0.0025	93.4	✓

The four Webb scenarios with <50% reduction all involve recoding concentration from dot to linear, which collapses two groups toward one another and reduces contrast power. Nonetheless the monotonic net>lin>dot ordering is preserved in every case. Full scenario-by-scenario results are reproduced by `evidence_synthesis/analysis/sensitivity_coding_audit.py`.

Greenwald topology sensitivity. The Greenwald topology analysis ($k = 184$ studies, four-level model) was tested under 31 alternative subdomain assignments involving five debatable subdomains. Across all 31 scenarios, the suppression effect at the linear-plus level was preserved (linear-plus

produced the lowest IEC in every case) and $\hat{\tau}^2$ reduction ranged from 28.7% to 34.7% (primary coding: 32.9%). Full scenario-by-scenario results are deposited at `evidence_synthesis/outputs/tables/greenwald2009_t`

Supplementary Figure 1: Cross-domain permutation distributions

Exhaustive permutation distributions of residual $\hat{\tau}^2$ for the three cross-domain datasets. Each histogram shows $\hat{\tau}^2$ for every unique surjective partition of the k units into the specified number of non-empty groups (Stirling number $S(n, k)$; group sizes free to vary). The dashed red line marks the DLN-coded $\hat{\tau}^2$. Numerical summaries are reported in Table 3 of the main text.

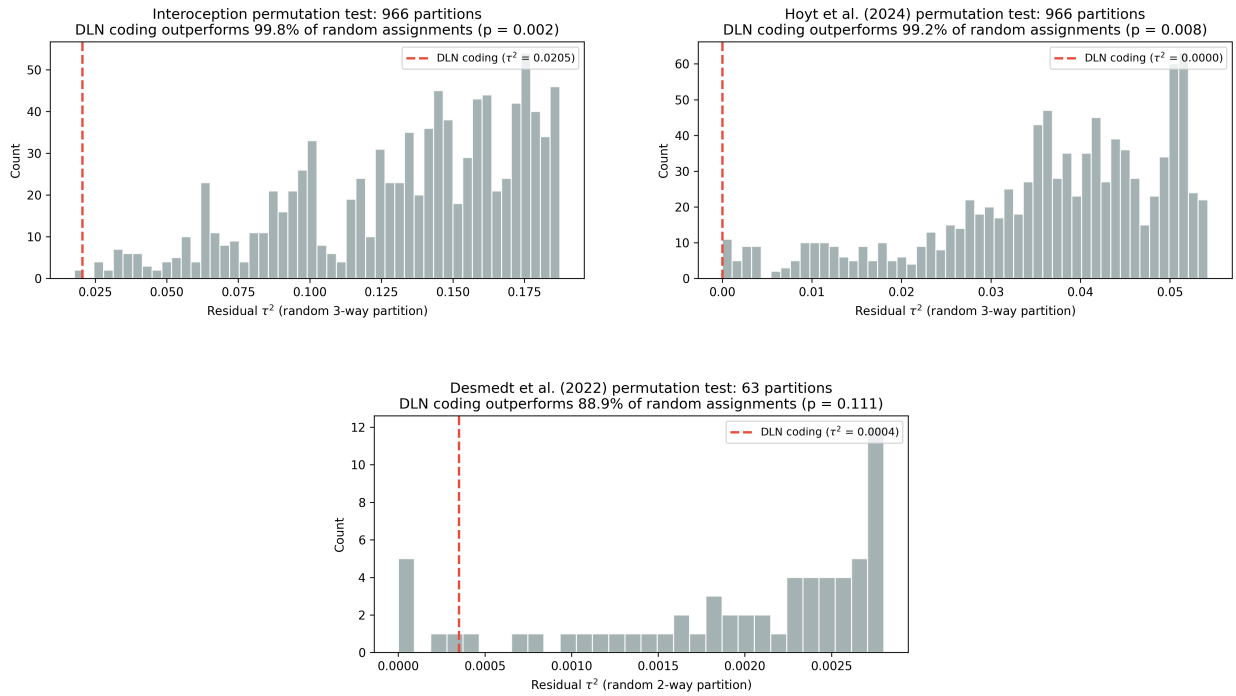

Figure 1: *

Supplementary Figure 1. Permutation distributions for cross-domain datasets. **(a)** Interoception–alexithymia ($k = 8$ measure families; $S(8, 3) = 966$ partitions; $p = 0.002$): DLN coding outperforms 99.8% of random three-way assignments. **(b)** Emotional approach coping and health (Hoyt et al., 2024; $k = 8$ outcome domains; $S(8, 3) = 966$; $p = 0.008$): DLN coding outperforms 99.2% of random assignments. **(c)** Heartbeat counting task criterion validity (Desmedt et al., 2022; $k = 7$ criteria; $S(7, 2) = 63$ two-way partitions; $p = 0.111$): DLN coding outperforms 88.9% of random assignments, though the test does not reach significance given limited combinatorial resolution (minimum achievable $p = 1/63 \approx 0.016$).

References

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